EE3110 - Probability Foundations for Electrical Engineers
Tutorial - Week 1

1. If n letters are placed at random in n envelopes, what is the probability that exactly $n-1$ letters will be placed in the correct envelopes?

   Solution:

   It is impossible to place exactly $n-1$ letters in the correct envelopes, because if $n-1$ letters are placed correctly, then the nth letter must also be placed correctly. Thus, it is a probability $\boxed{0}$ event.

2. An elevator in a building starts with five passengers and stops at seven floors. If every passenger is equally likely to get off at each floor and all the passengers leave independently of each other, what is the probability that no two passengers will get off at the same floor?

   Solution: There are 7^5 possible outcomes in the sample space. If the five passengers are to get off on different floors, the first passenger can get off on any one of the seven floors, and the second passenger can then get off on any one of the other six floors, etc. Thus, the probability is

   $$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^5} = \boxed{\frac{360}{2401}}.$$

3. Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.

   Solution: Define event G as the probability that a student is a genius, and event C as the probability that a student likes chocolate.

   We are given $P(G) = 0.6$, $P(C) = 0.7$, and $P(G \cap C) = 0.4$.

   We know that $P(G \cup C) = P(G) + P(C) - P(G \cap C) \Rightarrow P(G \cup C) = 0.9$. The probability that a randomly selected student is neither genius nor chocolate lover is $P(G' \cap C')$, by the De Morgan's law, $P(G' \cap C') = 1 - P(G \cup C) = \boxed{0.1}$.

4. You and your friend are playing a game with a standard deck of 52 playing cards. Each card type (2 through 10, Jack, Queen, King, Ace) appears exactly four times in the deck. For this problem, assume the following values for the cards:

   Number cards retain their face value (2 through 10)

   Jack = 11, Queen = 12, King = 13, Ace = 14

   You and your friend each draw one card from the deck without replacement. You draw first, followed by your friend. Whoever draws the card with the higher value wins. In case of a tie (both draw the same value), your friend wins.

   What is the probability that you win this game?

Solution: Method 1: By simple counting

Let us consider all the cards you could pick up. Each card has a $\frac{1}{13}$ chance of being picked.

- If you picked up a 2, you have $\frac{0}{51}$ chance of winning (since you chose the smallest card),
- If you picked up a 3, you have $\frac{4}{51}$ chance of winning (your friend picks a 2 among the remaining 51 cards),
- If you picked up a 4, you have $\frac{8}{51}$ chance of winning (your friend picks a 2 or a 3),
- $\vdots$
- If you picked up an Ace, you have $\frac{48}{51}$ chance of winning.

So your probability of winning is:

$$\frac{1}{13}\left(\frac{0+4+8+\cdots+48}{51}\right) = \frac{1}{13}\cdot\frac{4}{51}(0+1+2+\cdots+12) = \frac{4}{13\cdot 51}\cdot\frac{12\cdot 13}{2} = \frac{8}{17}$$

Method 2: By Symmetry

Define the following events:

- E_1: Your number is larger than your friend's.
- E_2: Your number is the same as your friend's.
- E_3: Your number is smaller than your friend's.

We want to find $P(E_1)$.

By symmetry, $P(E_1) = P(E_3)$, and by the law of total probability:

$$P(E_1) + P(E_2) + P(E_3) = 1 \Rightarrow 2P(E_1) + P(E_2) = 1$$

So, we just need to find $P(E_2)$.

Suppose you randomly pick a card. Among the remaining 51 cards, 3 of them have the same value as your card. Therefore:

$$P(E_2) = \frac{3}{51}$$

Substituting back:

$$2P(E_1) = 1 - \frac{3}{51} = \frac{48}{51} \Rightarrow P(E_1) = \frac{24}{51} = \boxed{\frac{8}{17}}$$

5. Let A and B be two events such that $\boldsymbol{P}(A) > \boldsymbol{P}(B) > 0$ and $\boldsymbol{P}(A) + \boldsymbol{P}(B) \geq 1$.
   (a) Determine upper and lower bounds on $\boldsymbol{P}(A \cap B)$ in terms of $\boldsymbol{P}(A), \boldsymbol{P}(B)$.
   (b) When $\boldsymbol{P}(A) \geq 1 - \delta$ and $\boldsymbol{P}(B) \geq 1 - \delta$, express the lower bound on $\boldsymbol{P}(A \cap B)$ in terms of δ.

Solution:

(a)

$$\boldsymbol{P}(A \cap B) = \boldsymbol{P}(A) + \boldsymbol{P}(B) - \boldsymbol{P}(A \cup B) \geq \boldsymbol{P}(A) + \boldsymbol{P}(B) - 1$$

$$(A \cap B) \subseteq A, (A \cap B) \subseteq B \Rightarrow \boldsymbol{P}(A \cap B) \leq min[\boldsymbol{P}(A), \boldsymbol{P}(B)]$$

(b)

$$\boldsymbol{P}(A \cap B) \geq \boldsymbol{P}(A) + \boldsymbol{P}(B) - 1 \geq 1 - \delta + 1 - \delta - 1 = \begin{cases} 1 - 2\delta & , if\ \delta \leq 1/2 \\ 0 & , \text{otherwise.} \end{cases}$$

6. There are 2^k tennis players. The players are of equal ability, i.e. in a match between any two players, probability of each of them winning is $\frac{1}{2}$.

   The players play an elimination tournament with random initial seeding. That is, the players are arranged randomly, then 2^{k-1} matches are constituted among these players such that seed 1 plays seed 2 in match 1, seed 3 plays seed 4 in match 2, and so on. The winners of these matches go to the next round which would have 2^{k-1} players and 2^{k-2} rounds (winner of match 1 plays winner of match 2), and so on until there is a single winner.

   1. What is the probability that any two given players meet in the first round?
   2. What is the probability that any two given players have a match during the final?
   3. What is the probability that any two given players have a match during the tournament?

Solution:

1. Without loss of generality, assume that player 1 is also seeded first. Now, there are $2^k - 1$ potential seedings for the second player, and all of them are equally likely. Thus, the desired probability is $\boxed{\frac{1}{2^k - 1}}$.
2. Every pair is equally likely to meet in the finals. There are $\binom{2^k}{2} = 2^{k-1} \cdot (2^k - 1)$ many possible pairs, and thus probability of a given pair meeting in the final is $\boxed{\frac{1}{2^{k-1} \cdot (2^k - 1)}}$.
3. The total number of possible games in the tournament is $2^k - 1$, and each of these games are equally likely. The total number of possible games is $\binom{2^k}{2}$. Thus, the probability that a particular pair has a match is just $\frac{2^k - 1}{2^{k-1}(2^k - 1)} = \boxed{\frac{1}{2^{k-1}}}$.

7. Sixteen players, $P_1, P_2, \ldots, P_{16}$, compete in a knockout tournament. It is known that whenever players P_i and P_j face each other, P_i will win if $i < j$. Assuming the players are paired randomly in each round, what is the probability that player P_4 reaches the semi-finals?

Solution: P_4 will reach the semifinal if and only if none of P_1, P_2, P_3 is in the same quarter of the final draw as P_4. The probability of this event E is,

$$P(E) = \frac{\binom{12}{3}}{\binom{15}{3}} = 44/91$$

8. You roll a fair 6-sided die n times. Find the probability that all 6 sides appear at least once. Also calculate the probability that exactly 4 of the 6 sides appear among the n rolls.

Solution: Let's look at what the complement of the required probability is. So the complement is the case where at least one of the faces do not appear after n rolls. Motivated by this, we'll define a few events.
Let B_1 denote the event where the 1- face does not appear even once among the n rolls of the die. Similarly define B_2, B_3 $\ldots B_6$.

The probability of the complement, therefore, is $\mathbb{P}(B_1 \cup B_2 \cup \cdots \cup B_6)$.

From the Inclusion-Exclusion Principle (IEP), we have

$$\mathbb{P}(\bigcup_{i=1}^{6} B_i) = \Sigma_{i=1}^{6}\mathbb{P}(B_i) - \Sigma_{i,j\in\{1,2,\ldots,6\}}\mathbb{P}(B_i \cap B_j) + \cdots - \mathbb{P}(B_1 \cap B_2 \cap \cdots \cap B_6).$$

By symmetry, $\mathbb{P}(B_i)$ is equal for all i, $\mathbb{P}(B_i \cap B_j)$ is the same for each of the $\binom{6}{2}$ pairs (i, j) chosen from $\{1, 2, \ldots, 6\}$ etc.,
We also have $\mathbb{P}(B_1)$ = Probability that the 1-face does not appear in any one of the n rolls = All rolls have faces chosen from the remaining 5 sides = $\frac{5^n}{6^n}$. Similarly,
$\mathbb{P}(B_1 \cap B_2) = \frac{4^n}{6^n}$ and so on. Plugging in the values, the required probability is 1 - Pr(complement) and is given by:

$$\boxed{1 - \binom{6}{1}\frac{5^n}{6^n} + \binom{6}{2}\frac{4^n}{6^n} - \binom{6}{3}\frac{3^n}{6^n} + \binom{6}{4}\frac{2^n}{6^n} - \binom{6}{5}\frac{1^n}{6^n}.}$$

The procedure is very similar for the second part of the question. Start by assuming that the 4 faces are $\{1, 2, 3, 4\}$. Then Pr(faces 1,2,3 and 4 appear at least once among n rolls and no other face does) is equal to

$$\frac{4^n}{6^n} - \binom{4}{1}\frac{3^n}{6^n} + \binom{4}{2}\frac{2^n}{6^n} - \binom{4}{3}\frac{1^n}{6^n}$$

Since the first 4-tuple could have been chosen in $\binom{6}{4}$ ways, the final answer is

$$\boxed{\binom{6}{4}\left(\frac{4^n}{6^n} - \binom{4}{1}\frac{3^n}{6^n} + \binom{4}{2}\frac{2^n}{6^n} - \binom{4}{3}\frac{1^n}{6^n}\right).}$$

9. Each of k jars contains m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally, a ball is randomly chosen from jar k. Show that the probability that the last ball is white is the same as the probability that the first ball is white, i.e., it is $m/(m + n)$.

Solution:

Intuitive Explanation: When a ball is picked up from the first jar, it has $\frac{m}{m+n}$ of being white. Now, when this ball is put in the next jar, the number of balls in the second jar increases by 1 but probability of picking a white ball remains the same. When a ball is picked up from the second jar, it again has a $\frac{m}{m+n}$ chance of being white, and when this ball is put in the third jar, this probability is maintained, and so on.

We derive a recursion for the probability p_i that a white ball is chosen from the i th jar. We have, using the total probability theorem,

$$p_{i+1} = \frac{m+1}{m+n+1}p_i + \frac{m}{m+n+1}(1-p_i) = \frac{1}{m+n+1}p_i + \frac{m}{m+n+1},$$

starting with the initial condition $p_1 = m/(m+n)$. Thus, we have

$$p_2 = \frac{1}{m+n+1}\cdot\frac{m}{m+n} + \frac{m}{m+n+1} = \frac{m}{m+n}.$$

More generally, this calculation shows that if $p_{i-1} = m/(m+n)$, then $p_i = m/(m+n)$. Thus, we obtain $p_i = m/(m+n)$ for all i.

10. Suppose that a box contains r red balls, w white balls, and b blue balls. Suppose also that balls are drawn from the box one at a time, at random, without replacement. What is the probability that all r red balls will be obtained before any white balls are obtained?

Solution: In the particular question asked here, blue balls are irrelevant. Since sequences we are interested in, the locations of blue balls do not matter.

Therefore, looking at only the red and white balls, there are $\binom{r+w}{r}$ possible positions that the red balls could occupy in the ordering as they are drawn. Therefore, the probability that they will be in the first r positions is $\frac{1}{\binom{r+w}{r}}$.

11. What is the smallest number k such that in a group of k people, there is a greater than 50% chance that at least two people share the same birthday? (Assume there are exactly 365 days in a year and each day is equally likely to be someone's birthday. Use the approximation $\ln(1-x) \approx -x$ for small x and $\ln(e^x) = x$)

Solution: First, we need to calculate the probability that no two people share the same birthday. Then, we will determine when this probability drops below 50%.

So, for k people, the probability $P(\text{no shared birthday})$ is given by:

$$P(\text{no shared birthday}) = \frac{365}{365}\times\frac{364}{365}\times\frac{363}{365}\times\cdots\times\frac{365-k+1}{365}$$

This can be simplified to:

$$P(\text{no shared birthday}) = \frac{365!}{(365-k)!\cdot 365^k} = \prod_{i=0}^{k-1}\left(1-\frac{i}{365}\right)$$

Using the Approximation $\ln(1-x) \approx -x$ for small x, we have the product as:

$$\ln\left(P(\text{no shared birthday})\right) = \sum_{i=0}^{k-1}\ln\left(1-\frac{i}{365}\right) \approx \sum_{i=0}^{k-1} -\frac{i}{365}$$

On simplifying :

$$P(\text{no shared birthday}) \approx \exp\left(-\frac{k(k-1)}{2\cdot 365}\right)$$

We want this to be less than 0.5; solving the quadratic equation gives us the least integer satisfying this is $k = 23$. It is non-trivial that with just 23 people in a room, there is over a 50% chance that at least two of them share the same birthday!

12. I bought two fair six-sided dice. Keeping the first one untouched (numbered 1 through 6), I modified the second die by erasing all its faces and labeling them with real numbers of my choice (not necessarily distinct). After rolling both dice together, I observed that the sum of the two outcomes is equally likely to be any integer from 1 to 12. Determine the numbers written on the modified die.

Solution: Let D_1 be the fair standard die with faces $\{1, 2, 3, 4, 5, 6\}$, and let D_2 be the modified die with faces $\{x_1, x_2, x_3, x_4, x_5, x_6\}$. We are told that the sum $S = D_1 + D_2$ is uniformly distributed over $\{1, 2, \ldots, 12\}$, i.e.,

$$\mathbb{P}(S = k) = \frac{1}{12} \quad \text{for all } k \in \{1, 2, \ldots, 12\}.$$

Since the minimum and maximum possible sums are 1 and 12 respectively, we must have:

$$S = 1 \iff D_1 \text{ has 1, } D_2 \text{ has 0,} \quad \text{and} \quad S = 12 \iff D_1 \text{ has 6, } D_2 \text{ has 6.}$$

Hence, the modified die must include the values 0 and 6. Furthermore, to ensure the sum lies within $\{1, 2, \ldots, 12\}$, each face of D_2 must take a value from the set $\{0, 1, 2, 3, 4, 5, 6\}$.

Let R and M denote the outcomes of D_1 and D_2, respectively. Then,

$$\begin{aligned}\mathbb{P}(R + M = 1) &= \mathbb{P}(R = 1 \wedge M = 0) \\ &= \mathbb{P}(R = 1) \cdot \mathbb{P}(M = 0) = \frac{1}{6} \cdot \mathbb{P}(M = 0).\end{aligned}$$

Since this must equal $\frac{1}{12}$, we get:

$$\frac{1}{12} = \frac{1}{6} \cdot \mathbb{P}(M = 0) \Rightarrow \mathbb{P}(M = 0) = \frac{1}{2}.$$

Similarly, for $S = 12$:

$$\begin{aligned}\mathbb{P}(R + M = 12) &= \mathbb{P}(R = 6 \wedge M = 6) \\ &= \frac{1}{6} \cdot \mathbb{P}(M = 6) = \frac{1}{12},\end{aligned}$$

which implies:

$$\mathbb{P}(M = 6) = \frac{1}{2}.$$

As the probabilities for all other face values must sum to zero, no other number can appear on D_2. Since D_2 has six faces and both 0 and 6 occur with equal probability, it must have three faces labeled 0 and three faces labeled 6.

Thus, the modified die has the face values: $\{0, 0, 0, 6, 6, 6\}$.

13. **Circular Equilibrium and Stability:**
A circular platform floats frictionlessly in space. You place N identical point masses on the circumference of the platform. Each mass is placed independently and uniformly at random. The system is said to be in a *stable configuration* if there exists a rotation of the platform such that all the masses lie within a semicircle (i.e., within an arc of length π). What is the probability that the configuration is stable?

Solution: Let the N points be labeled $P_1, P_2, \ldots, P_N$, placed independently and uniformly at random on the circumference of a circle.

We aim to compute the probability that *all points lie within some semicircle.*

Fix an arbitrary point P_i and rotate the circle such that it lies at angle 0. We now ask: what is the probability that all the remaining $N-1$ points lie within the clockwise semicircle of length π (i.e., the upper half-plane from angle 0 to π)?

Since the points are placed independently and uniformly at random, each of the remaining $N-1$ points has a probability $\frac{1}{2}$ of falling in this semicircle. Therefore, the probability that all of them do is:

$$p = \left(\frac{1}{2}\right)^{N-1}$$

Now observe that this configuration - where all points lie in a semicircle with one designated point as the leading point can occur for any of the N points. Moreover, the events that each point is the leading point are *mutually exclusive*: if P_i is the unique point such that all others lie within its semicircle, then no other point can satisfy this condition simultaneously.

$$\mathbb{P}(\text{All points lie within some semicircle}) = \sum_{i=1}^{N} \mathbb{P}(E_i) = N \cdot \left(\frac{1}{2}\right)^{N-1} = \boxed{\frac{N}{2^{N-1}}}$$

Sanity Check:

- For $N = 1$: Clearly the point lies in a semicircle. Probability $= \frac{1}{2^0} = 1$.
- For $N = 2$: Any two points lie in some semicircle. Probability $= \frac{2}{2^1} = 1$.

14. **(Something to think about)** Assume that there is a roulette wheel, such that it is equally likely to land at any particular angle. The probability of getting a particular angle (say 30°) is 0. However, once the roulette wheel is spun, it would have to land somewhere (say it lands at 30°). Thus, a probability 0 event has occurred. Is this a discrepancy?

Solution: There is no discrepancy. Events with zero probability can still occur in continuous distributions.